GENE STORM
Game Design Document

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Contents

1 Overview

1.1 Summary

1.2 Vision

1.3 Scope

1.4 Glossary

2 Core Loop

2.1 Game Progression

2.2 Player Interaction

2.3 Reward an Motivations

2.4 Content Roadmap

3 System Specifications

4 Stats and Formulas

5 Fighting System

6 Skill System

7 Items and Inventory

8 Crafting System

8.1 Gear Crafting

8.1.0.1 Overview

The Gear Crafting System combines deterministic player-driven modification (Affixes) with random initial base generation (Implicits). Each gear item is defined by its **Gear Type** (Head, Chest, Leg, Foot, Gloves, Amulet) and can contain:

- Up to **N Implicits** (rolled automatically on creation)
- Up to **M Affixes** (player-crafted and upgradable)

Each mechanic type has its own input/output schema, cost structure, and scaling formula. All crafting transactions are atomic (no partial resource loss).

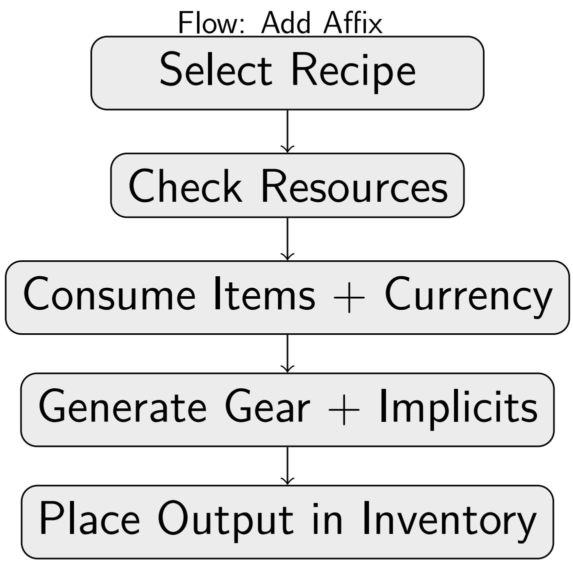
Design Goal: Create meaningful player agency through controlled randomness and resource-driven progression.

8.1.1 Mechanic I – Basic Crafting (Gear Recipes)

8.1.1.1 Purpose

Convert raw materials (and optional boosters) into finished base gear items.

8.1.1.2 Sequence



8.1.1.3 Input / Output Schema

Element	Description / Constraints
Ingredients	Up to 4 material items (each with quantity).
Booster	Max 1 optional booster item (may modify tier/roll bias).
Currency	Any number of currency types (wallet-based).
Output	1 gear item (ItemBaseDef reference).

8.1.1.4 Cost Structure

- Base cost determined by item rarity and recipe complexity.
- Optional booster adds both cost and potential reward.
- Currency cost acts as sink to regulate player economy.

8.1.1.4.1 Cost Scaling

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8.1.1.5 Booster Effect

based on Rarity/Qualtiy and Quantity of the used Booster-Material the output is modified.

Parameter	Symbol / Range	Description
Tier Boost per Unit	$f_1(q)$	Adds virtual tier levels to implicits.
Roll Bias per Unit	$f_2(q)$	Shifts random roll toward max value.
Caps	$\mathrm{cap}_T, \mathrm{cap}_B$	Hard caps for both effects.
Diminishing Curve	$D(q)$	Function for diminishing returns.

8.1.1.6 Implicits

Implicits and base rolls per Tier

Kurzname	Implicit Effekt	Type	Tag	T1 roll	T2 roll	T3 roll
DMG%	increased damage	%	damage	5–10	10–15	15–25
Thorns	to thorns	flat	damage	1–5	5–7	7–12
Phys+	to phys damage	flat	damage	1–5	5–7	7–12
Elem+	to elemental damage	flat	damage	1–5	5–7	7–12
Armor%	increased armor	%	defence	5–10	10–15	15–25
Res%	increased elemental resistance	%	defence	5–10	10–15	15–25
Dodge%	increased dodge chance	%	defence	5–10	10–15	15–25
Barrier	increased barrier	%	defence	5–10	10–15	15–25
Armor+	to armor	flat	defence	30–60	60–100	100–150
MaxBarrier+	to max barrier	flat	defence	30–60	60–100	100–150
Life%	increased life	%	life	5–10	10–15	15–25
MaxLife+	to max Life	flat	life	20–40	40–75	75–120
Rarity%	increased item rarity	%		5–7	7–10	10–15
Speed%	increased movement speed	%	speed	10–15	15–25	25–40

weight rolls of implicit for each geartype

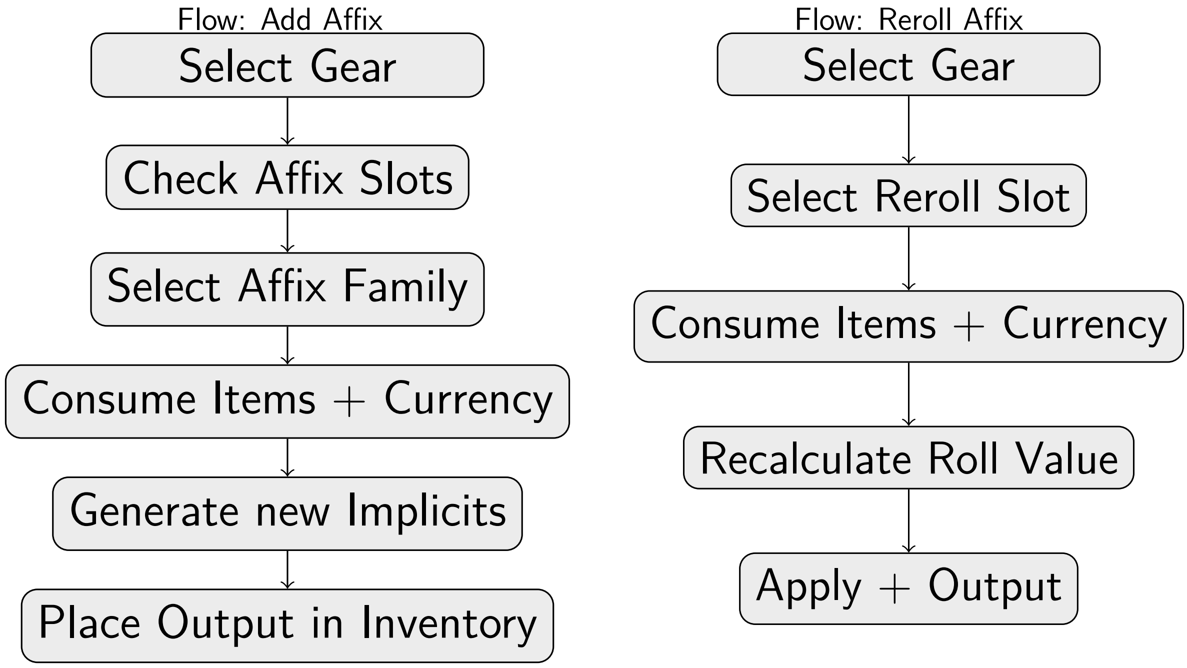
Stat	Head	Chest	Gloves	Legs	Boots	Amulet
DMG%	100	50	300	100	50	500
Thorns	0	100	50	100	300	0
Phys+	100	50	300	100	50	300
Elem+	300	50	100	100	50	300
Armor%	100	500	50	300	100	100
Res%	300	500	0	300	50	100
Dodge%	300	100	50	500	200	50
Barrier%	300	300	100	100	50	200
Armor+	100	500	50	100	100	50
MaxBarrier+	100	300	100	100	100	50
Life%	100	500	50	100	50	200
MaxLife+	100	500	50	100	100	50
Rarity%	50	25	50	50	100	100
Speed%	0	0	0	0	750	0
sum	1950	3475	1250	2050	2050	2000

8.1.2 Mechanic II – Upgrade / Affix Crafting

8.1.2.1 Purpose

Allows the player to add new affixes, upgrade existing tiers, or perform one reroll on a designated slot.

8.1.2.2 Sequence



8.1.2.3 Input / Output Schema

Element	Description / Constraints
Base Item	Gear item with available affix slots.
Materials	Crafting reagents depending on action.
Currency	Gold or rare currencies.
Output	Mutated item (same ID, new affix or tier).

8.1.2.4 Cost Structure

- **Add Affix:** base cost plus weighted RNG PickSize (2–4).
- **Reroll:** 1 time per item: high base csot, escalating with tier.

8.1.2.4.1 Cost Scaling

tbd

8.1.2.5 Affix Pools

8.1.2.5.1 Core Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
to STR	flat	attribute	3–7	7–15	15–20
to DEX	flat	attribute	3–7	7–15	15–20
to CON	flat	attribute	3–7	7–15	15–20
to INT	flat	attribute	3–7	7–15	15–20
to WIL	flat	attribute	3–7	7–15	15–20
increased crit chance	%	crit	10–15	15–25	25–40
increased crit dmg	%	crit	30–60	60–100	100–150
increased phys damage	%	damage	20–40	40–75	75–120
increased elemental damage	%	damage	20–40	40–75	75–120

8.1.2.5.2 Defensive Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
increased all resistance (phys/elem)	%	defence	10–15	15–25	25–40
reduced DoT duration	%	defence	10–15	15–25	25–40
gain % reflect equal to dodge rating	%	defence	20–40	40–75	75–120
to barrier regen per second	flat	defence	3–7	7–15	15–20
reduce barrier recharge delay	%	defence	10–15	15–25	25–40
% dmg taken recouped as life	%	life	1–4	4–8	8–12
leech damage as life	%	life	1–4	4–8	8–12
to life regen per second	flat	life	5–10	10–25	25–40

8.1.2.5.3 Synergy Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
% armor applies to all Dmg	%	damage	10–15	15–25	25–40
% crit chance applies to phys Dmg	%	damage	10–15	15–25	25–40
% dodge rating applies to thorns	%	damage	10–15	15–25	25–40
% Barrier applies to elem damage	%	damage	10–15	15–25	25–40
% phys damage penetrates armor	%	damage	10–15	15–25	25–40
% elem damage penetrates resistance	%	damage	10–15	15–25	25–40

8.1.2.5.4 Utility Pool

Affix effect	type	tag	T1 roll	T2 roll	T3 roll
increased melee skill power	%	skill	20–40	40–75	75–120
increased projectile speed	%	skill	20–40	40–75	75–120
increased area of effect	%	skill	20–40	40–75	75–120
increased ranged skill power	%	skill	20–40	40–75	75–120
increased range	%	skill	20–40	40–75	75–120
increased pierce chance	%	skill	20–40	40–75	75–120
increased cast speed	%	speed	20–40	40–75	75–120
reduced cooldown	%	speed	10–15	15–25	25–40
increased item rarity	%		10–15	15–25	25–40

8.1.2.6 Reroll Affix

- Only one slot can be rerolled.
- Rerolls the Affix effect
- no duplicate of existent Affixes
- the more Affixes are on the item, the smaller the Reroll-Range of which Affix will be rolled
- relativley high base cost, but no Sclaing, since 1 time per item

8.1.3 Mechanic III – Hardening (Tier Enhancement)

8.1.3.1 Purpose

A late-game mechanic that permanently increases item potential through tier reinforcement.

8.1.3.2 sequence

Select Hardenable Item
↓
Check Max Affix Tier
↓
Consume High-tier Materials
↓
Apply Hardened Level +1
↓
Place Back in Inventory

8.1.3.3 Input / Output Schema

Element	Description / Constraints
Base Item	Crafted item with maxed affixes.
Materials	Rare tier-enhancing reagents.
Currency	High-end currency cost (flat + exponential).
Output	Same item with upgraded "Hardened" property.

8.1.3.4 Hardening Tiers

Level	Cost Multiplier	Effect	Notes
1	x1	increase roll within the tier	
2	x2	increase roll within the tier	
3	x4	+1 Tier	
4	x5	increase roll within the tier	
5	x6	increase roll within the tier	
6	x12	+1 Tier	
7	x13	increase roll within the tier	
8	x14	increase roll within the tier	
9	x28	+1 Tier	
10	x29	increase roll within the tier	
11	x30	increase roll within the tier	
12	x60	Overroll (over Max)	extreme high cost for alte endgame

8.1.3.5 Cost Structure

- Each hardening level multiplies base material and currency cost.
- Rare reagents required (e.g., Essence, Soul Alloy).

8.1.3.5.1 Cost Scaling

see Cost Multiplicator in Hardening Tiers

8.1.4 General Validation Rules

- Recipes must define ≤ 4 ingredients and ≤ 1 booster.
- All costs non-negative; zero costs invalid.

- ItemBaseDef must specify GearType and AllowedPools.
- Atomic transactions: no partial resource consumption.
- Rollback mandatory on any failure path.

9 Perks and Traits

9.1 Loot and Rarities

Purpose

This section specifies the complete loot flow from wave setup to final rewards. It is implementation-agnostic and defines inputs, order of operations, and balancing knobs.

Key Terms

- **Wave Budget** B: Total loot value budget allocated to a wave.
- **Spent Budget** U: Accumulated loot value already awarded in the current wave.
- **Rarity** R: Item quality tier (e.g., Common, Uncommon, Rare, Epic, Legendary, $\ldots$).
- **Base Chance** $p_{\textbf{base}}[R]$: The configured per–rarity drop chance before any modifiers.
- **Unlucky Multiplier** $m_{\textbf{unlucky}}[R]$: Pity-protection factor derived from recent fails / hits per rarity.
- **Luck Factor** L: Linear luck strength; $L = 0$ (no bonus), $L = 1$ ($+100$
- **Luck Scale** $k[R]$: Optional per–rarity scaling of luck (default $k[R] = 1.0$).
- **Budget Multiplier** $m_{\textbf{budget}}(U, \textbf{lootValue}, B, R)$: Soft damping when U approaches/exceeds B, including rarity floors.
- **Effective Tags**: The union of an enemy's *baseTags* and *bonusTags*.

0) Wave Setup

1. **Compute Wave Budget** B: Derived from wave composition (ranks, counts), map difficulty and level; add variance; initialize $U = 0$.
2. **Load and Merge Loot Rules**: For all tags present in the wave, prepare the merged rule set (drop pools, min/max drops, rarity guarantees, secret rules).

1) On Enemy Kill

1. **Determine Roll Count**: Based on enemy rank/type (e.g., Elite may trigger more rolls than Normal).
2. **Build Effective Tag Set**: $effectiveTags = baseTags \cup bonusTags$ (duplicates ignored for logic; weighting optional).

2) Per Roll (for this enemy)

1. **Pick Tag**: Randomly select one tag from $effectiveTags$ (optionally bias bonusTags slightly).
2. **Resolve Tag Rule**: Activate the corresponding merged loot rule for this tag (yields a list of loot entries to attempt).

3) Iterate the Tag's Drop Entries

For each *loot entry* (which may be a single chance or a list of chances), evaluate one *rarity block* at a time. The system supports *parallel rarity rolls* so that multiple rarities can succeed in the same slot.

Factor Order (per rarity R):

1. **Base Chance**: $p_{\text{base}}[R]$
2. **Unlucky (Pity)**: multiply by $m_{\text{unlucky}}[R]$
3. **Luck (Linear, No Cap)**: multiply by $(1 + L \cdot k[R])$ *Example:* $p_{\text{eff, preBudget}}[R] = p_{\text{base}}[R] \cdot m_{\text{unlucky}}[R] \cdot (1 + L \cdot k[R])$
4. **Budget (Soft Brake + Floors)**: multiply by $m_{\text{budget}}(U, \text{lootValue}, B, R) \to p_{\text{roll}}[R] = p_{\text{eff, preBudget}}[R] \cdot m_{\text{budget}}(\cdot)$
5. **RNG Roll**: If success, **award one item** of rarity R (this slot can yield multiple items if several rarities succeed).

Post-Roll Accounting (per rarity):

- On success: add the item to results; increase U by its `lootValue`; call Unlucky.OnDrop(R).
- On fail: call Unlucky.OnFail(R).

4) Secret Drops (Optional, Per Enemy)

Evaluate secret/conditional drops (e.g., specific tag combinations). These are additional rolls that can yield extra items. They do not interfere with the regular pity counters unless explicitly specified.

5) Wave Finalization

1. **MinDrops Failsafe**: If total items $<$ configured minimum for the wave, add items from the wave's mixed pool until the minimum is met. This path increases U; whether it resets pity per rarity is a policy toggle (recommended: *does not* reset pity to preserve fairness).
2. **Rarity Guarantees**: If a minimum per rarity is required, top up those rarities. This path typically *does not* alter pity and may optionally bypass budget damping depending on economy policy.

Design Notes

- **Parallel Rarity Rolls**: Each rarity is evaluated independently; *all successes drop*. Lower-tier success does not consume or block higher-tier success.
- **Luck Scope**: Luck affects *quality* (rarity chances) only. Quantity remains governed by entries, rules, and budget. If desired, allow certain luck-driven drops to *ignore* budget damping (economy-sensitive).
- **Budget as Soft Brake**: m_{budget} smoothly reduces chances when U exceeds B; rarity floors prevent near-zero results for higher tiers.
- **Tags as Theme Driver**: Use $effectiveTags$ for both regular and secret rolls; optional small biases can keep bonusTags slightly more impactful without excluding baseTags.

Balancing Knobs

- **Per–rarity base chances** $p_{\text{base}}[R]$
- **Unlucky growth/reset rates** per rarity
- **Luck factor** L (from passive tree, map mods, etc.) and **luck scales** $k[R]$
- **Budget damping** curve parameters and **rarity floors**
- **Tag pick weighting** (base vs. bonus) and secret-drop criteria
- **Failsafe policies**: whether MinDrops/Guarantees affect pity or budget

9.2 Meta Progression

10 Content Bible

10.1 Archetypes and Characters

10.2 Skill Catalog

10.3 Modificator Catalog

10.4 Enemy Taxonomy

11 Technical Design

11.1 Architecture Overview

11.2 Datamodell

11.3 Runtime-Pipelines

11.4 Saving

11.5 Tooling and Debugging

11.6 Performance Budget

11.7 Coding Guidelines

12 UI/UX Specification

12.1 Information Flow

12.2 Key Screens

12.3 Input Periphery

12.4 Feedback

13 Balance and Tuning

13.1 Target Values

13.2 Curves

13.3 Test Scenarios